Appendices A and B: the models of the di-Higgs m_{hh} spectrum

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A The model of the m_{hh} spectrum

This appendix describes the model of the gluon-fusion di-Higgs spectrum used in this paper: how it is built and fitted to the POWHEG spectra, how its uncertainty is obtained, including a systematic uncertainty, and how both are validated on spectra not used in the fit. A second, independent model of the same quantity is described in appendix B.

A.1 The form of the cross section and the data

For the HEFT parameters $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$ the differential cross section at NLO in QCD has the form [1, 2]

$$\frac{d\sigma}{dm_{hh}}(m_{hh};\theta) = \sum_{i=1}^{23} C_i(\theta)\, a_i(m_{hh}), \tag{1}$$

where the C_i are known monomials in θ (products of the coupling factors of the amplitude) and the coefficient functions a_i depend on m_{hh} only. Equation (1) is exact provided that nothing else in the calculation depends on θ (scales, parton distributions and cuts are the same for every θ). The task is to determine the 23 functions $a_i(m_{hh})$.

The data are POWHEG spectra at $\sqrt{s} = 13$ TeV, each at a different point θ in bins of 15 GeV, of which we use the 80 bins between $m_{hh} = 255$ and 1455 GeV. Each spectrum is a histogram of 40,000 generated events of equal weight q_s (a few incomplete runs have fewer), so that the cross section in bin k is exactly $t_{sk} = q_s n_{sk}$ with an integer count n_{sk}. We recover q_s and the counts of every spectrum as the greatest common divisor of its bin contents. The counts are Poisson distributed and, up to a negligible multinomial correlation, independent; near threshold a bin holds about 1,000–1,800 events, at 1.4 TeV about 20. One thousand spectra are set aside as *test spectra* and play no part in any fit. The bins of the remaining 3,989 spectra are split at random: 88% (281,044 bins, including 1,814 empty ones) are used in the fit, and the other 12% (38,076 bins, the *held-out bins*) are used only to determine the systematic uncertainty of section A.4.

A.2 The model

Each coefficient function is expanded in $J = 40$ cubic B-splines [3] B_j and multiplied by the two-body threshold factor ϕ,

$$a_i(m) = \phi(m) \sum_{j=1}^{J} \beta_{ij} B_j(m), \qquad \phi(m) = \sqrt{1 - \frac{4m_h^2}{m^2}}, \tag{2}$$

with $m_h = 125$ GeV. The factor ϕ, the velocity of the Higgs bosons in their rest frame, makes the density vanish at threshold as two-body phase space does, so that the splines describe only

the smooth remainder. The 38 interior breakpoints are uniform in $\sqrt{m - 2m_h}$ between 250 and 1500 GeV, and so are densest just above threshold, where the spectra change fastest (figure 1). The prediction for a bin is the *integral* of eq. (1) over the bin, not its value at the bin centre:

$$f_{sk} = \int_{\text{bin } k} \frac{d\sigma}{dm}(m;\theta_s)\, dm = \sum_{i=1}^{23} \sum_{j=1}^{J} C_i(\theta_s)\, \beta_{ij}\, I_{jk}, \qquad I_{jk} = \int_{\text{bin } k} \phi(m) B_j(m)\, dm. \tag{3}$$

The model is therefore a genuine density in m_{hh}, and it is *linear* in its $23J = 920$ parameters β_{ij}. The integrals I_{jk} are computed once by Gauss–Legendre quadrature. With one box-shaped basis function per bin in place of the splines, eq. (3) becomes a fit of 23 independent numbers per bin. This *free fit* describes the data as well as any model of the form (1) can, and serves as the benchmark below.

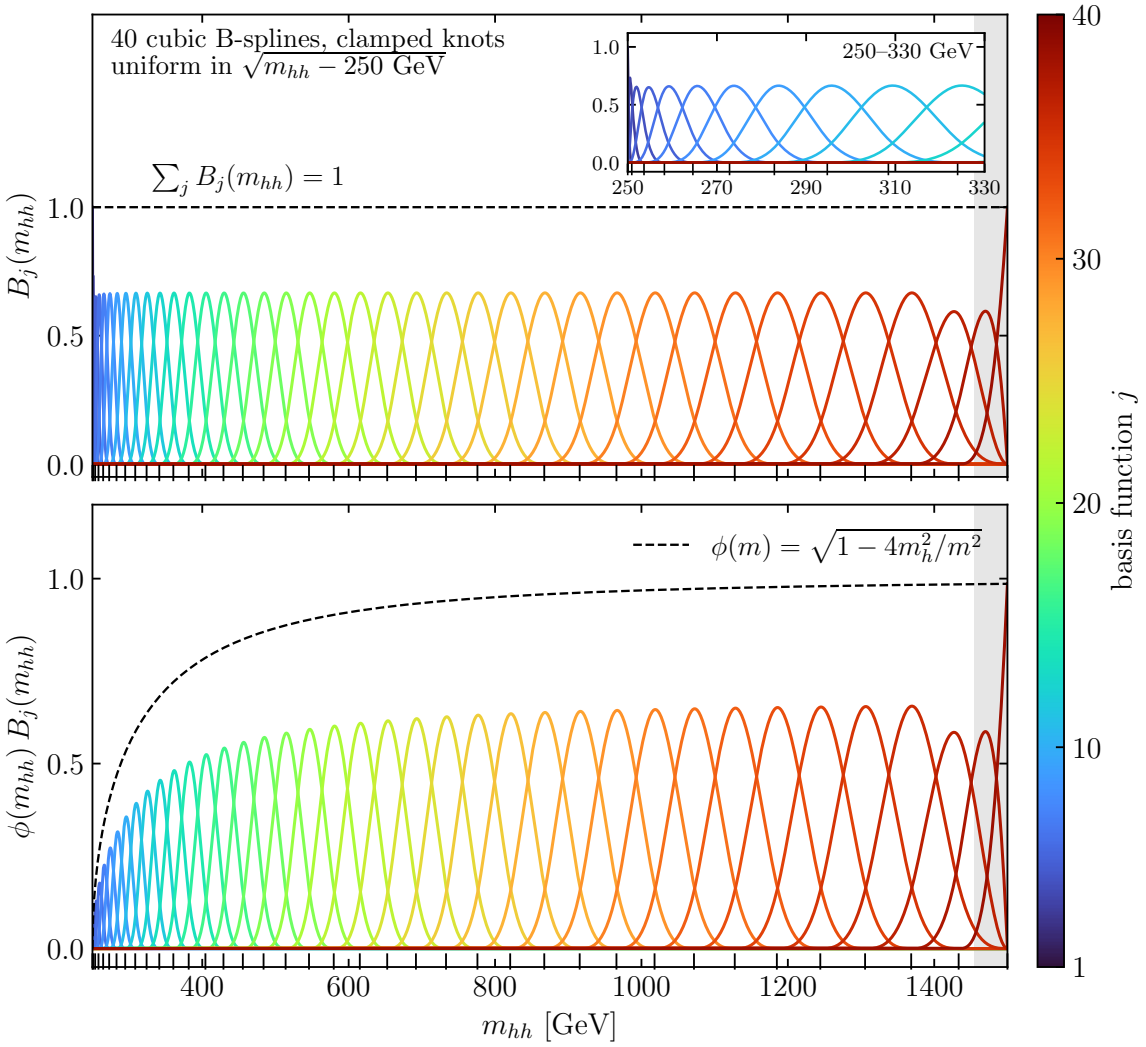


Figure 1: The basis of the model. Top: the 40 cubic B-splines $B_j(m_{hh})$, coloured by index j, with their breakpoints (ticks), uniform in $\sqrt{m_{hh} - 250\ \text{GeV}}$; the inset shows 250–330 GeV. The splines sum to one everywhere. Bottom: the products $\phi(m_{hh})B_j(m_{hh})$ that multiply the coefficients β_{ij}, with the threshold factor ϕ (dashed). The shaded region lies beyond the last bin used.

A.3 Fit and posterior

Likelihood. With hundreds of events in most bins a Gaussian likelihood with the Poisson variance is accurate, but the variance must be that of the model's prediction, $q_s \max(f_{sk}, q_s)$, not that of the observed count. Weighting each bin by $1/(q_s n_{sk})$ (Neyman's χ^2) gives downward fluctuations larger weights and biases the fit low by about one event per bin [4], which, with thousands of spectra pooled, is many standard deviations at high mass. We therefore iterate: fit, recompute the variances from the fit, and refit [5]. At convergence this is the Poisson maximum-likelihood fit.

Smoothness prior. For each function we use a Gaussian prior on the second differences of its coefficients [6],

$$-\ln \pi(\boldsymbol{\beta}_i \mid \lambda_i) = \tfrac{1}{2}\lambda_i \sum_{j=3}^{J} \left(\beta_{ij} - 2\beta_{i,j-1} + \beta_{i,j-2}\right)^2 + \text{const}, \tag{4}$$

plus a negligible ridge term, the discrete analogue of a penalty on the curvature $\int (a_i'')^2\, dm$. The 23 strengths λ_i are chosen by maximising the evidence [7], which is available in closed form because the model is linear.

Posterior. Likelihood and prior being Gaussian in $\boldsymbol{\beta}$, the posterior is *exactly* Gaussian, with mean $\hat{\boldsymbol{\beta}} = H^{-1}\Phi^{\mathsf{T}} W \boldsymbol{t}$ and covariance H^{-1}, where $H = \Phi^{\mathsf{T}} W \Phi + \Lambda$, Φ is the design matrix of eq. (3), W the diagonal matrix of inverse variances and Λ the prior precision. Because the monomials and the bins factorise, $\Phi^{\mathsf{T}} W \Phi = \sum_k S_k \otimes \boldsymbol{I}_k \boldsymbol{I}_k^{\mathsf{T}}$ with $S_k = \sum_s w_{sk} \boldsymbol{C}_s \boldsymbol{C}_s^{\mathsf{T}}$, and the whole fit takes about a minute on a laptop. The data determine about 600 of the 920 parameter combinations; the prior fixes the rest. Predictions for any binning, the density at any m_{hh}, the coefficient functions and the correlations between bins all follow from $\hat{\boldsymbol{\beta}}$ and H^{-1}. The *fit uncertainty* of a bin is small, a median 2.9% of the Poisson noise of a POWHEG bin. Nothing changes, within 0.01 in any measure below, for $J = 30$ or 60.

A.4 Systematic uncertainty: the model-error term

The comparison of a prediction f_{sk} with a POWHEG bin t_{sk} involves three uncertainties: the Poisson noise of the POWHEG sample, $s^2_{\text{noise}} = q_s \max(f_{sk}, q_s)$; the fit uncertainty of the model, s^2_{fit}; and a third, systematic uncertainty, which arises not from statistical fluctuations but from the fact that the POWHEG spectra do not exactly obey eq. (1). We describe the evidence for it, its form, and how it is determined. It measures the internal consistency of the simulated spectra; the uncertainties of the NLO prediction itself (scales, parton distributions, the top-mass scheme) are separate and are not included.

The evidence. The accuracy of a prediction $\mu = f/q$ for a count $n = t/q$ is measured by the likelihood-ratio statistic

$$-2\ln\lambda = -2\ln\frac{\text{Poisson}(n;\mu)}{\text{Poisson}(n;n)} = 2\left[\mu - n + n\ln\frac{n}{\mu}\right], \tag{5}$$

which for a correct prediction follows, asymptotically, a χ^2 distribution with one degree of freedom [8], with mean 1; no parameter is adjusted to the test spectra. Below 555 GeV its mean over the test spectra is 1.57, for the smooth model and for the free fit alike (table 1). Since the free fit is the best any model of the form (1) can do, the excess over 1 is a property of the spectra, not of the model. It depends on how strongly the 23 terms cancel. With the *cancellation ratio* of a spectrum, $R_s = \sum_k t_{sk} / \sum_k \sum_i |C_i(\theta_s) A_{ik}|$, where A_{ik} is the integral

of a_i over bin k, the fifth of the test spectra with the strongest cancellation ($R < 0.12$) has $\langle z^2 \rangle = 1.70$ for the pulls with noise and fit uncertainty only, against 1.01 for the fifth with the weakest ($R > 0.73$). In the strongest-cancelling fifth the residuals of neighbouring bins are correlated (0.60 below 555 GeV), so the departures are smooth in m_{hh}; Poisson noise would give no correlation.

The form. Suppose that in the spectrum of run s each of the 23 terms carries a small relative error $\delta_i(m; s)$. The error of the spectrum, $\sum_i C_i a_i \delta_i$, is then of order $\varepsilon \sum_i |C_i a_i|$ for errors of typical size ε, while the cross section is $\sum_i C_i a_i$: the relative error is amplified by $1/R$ where the terms cancel. The errors must also differ from run to run, since errors common to all runs would be absorbed into the a_i. We therefore add to the variance of each predicted bin the *model-error term*

$$s^2_{\text{model}} = \left(\varepsilon(m_k) \sum_{i=1}^{23} \left| C_i(\theta)\, A_{ik} \right| \right)^2, \tag{6}$$

in the spirit of the model discrepancy of Kennedy and O'Hagan [9]. By the triangle inequality, eq. (6) is the largest spread that relative errors of size ε can produce, whatever their correlation; for independent errors the sum would be taken in quadrature. Both forms calibrate the test spectra equally well (table 2); we adopt the more conservative one. A systematic proportional to the Poisson noise instead calibrates the average but not the cancelling spectra.

Determination. In each bin, ε_k is fixed on the held-out bins by requiring the pulls $z = (t - f)/\sqrt{s^2_{\text{noise}} + s^2_{\text{fit}} + s^2_{\text{model}}}$ to have unit variance; $\ln \varepsilon$ is then smoothed by a quadratic in m_{hh}, so that the term has three parameters in all (figure 2). It rises from 0.08% at threshold to 0.28% at 600 GeV and 3.7% at 1.45 TeV: near threshold the 23 components of the POWHEG spectra are consistent to about one part in a thousand. The model-error term is treated as independent between bins; the residual correlations above show that this is an approximation, which matters for sums over many bins (section A.7).

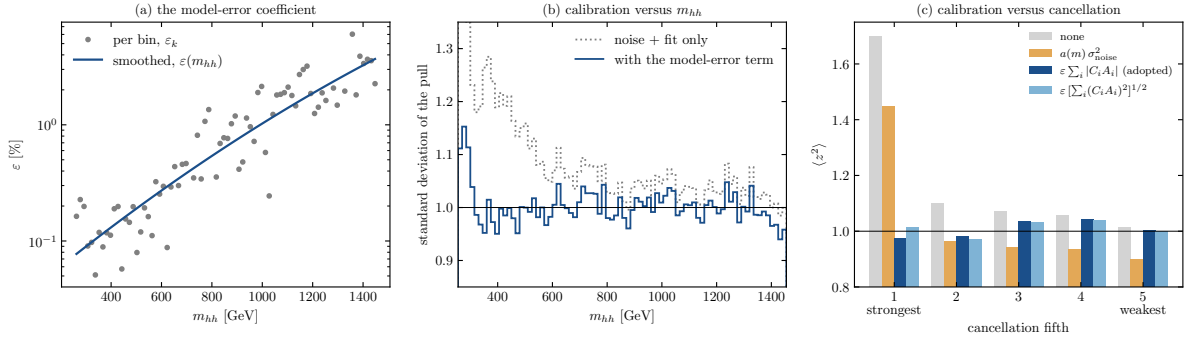


Figure 2: The model-error term. (a) The coefficient ε: per bin, from the held-out bins (points), and its three-parameter smooth form (line). (b) Standard deviation of the pull on the 1,000 test spectra in each bin, without (dotted) and with (solid) the model-error term. (c) $\langle z^2 \rangle$ by fifth of the cancellation ratio, for no systematic, a systematic proportional to the Poisson noise, and the model-error term summed linearly (eq. (6), adopted) and in quadrature. Each was determined on the held-out bins.

A.5 Validation on the test spectra

Accuracy. The smooth model reaches the floor set by the free fit: the mean $-2 \ln \lambda$ per bin is 1.566 below and 1.100 above 555 GeV (table 1, figure 3). The excess over 1 is the

	mean $-2\ln\lambda$ per bin		median
	255–555 GeV	>555 GeV	per spectrum
smooth model	1.566	1.100	1.078
free fit (benchmark)	1.559	1.105	1.075

Table 1: Accuracy on the 1,000 test spectra: the mean of the likelihood-ratio statistic (5) per bin, which is 1 for a perfect model, and the median over spectra of its mean per bin.

	s.d. of the pull				$\langle z^2\rangle$ by fifth of cancellation
systematic	all	255–555	>555 GeV	$\lvert z\rvert<2$	(strongest … weakest)
none	1.090	1.218	1.043	93.9%	1.70, 1.10, 1.07, 1.06, 1.01
proportional to the noise	1.018	1.049	1.008	95.3%	1.45, 0.96, 0.94, 0.94, 0.90
model-error term, eq. (6)	1.004	1.011	1.001	95.4%	0.97, 0.98, 1.04, 1.04, 1.00
the same, summed in quadrature	1.005	1.011	1.003	95.4%	1.01, 0.97, 1.03, 1.04, 1.00

Table 2: Calibration on the test spectra. For a standard normal, $|z|<2$ holds for 95.4%. Each systematic was determined on the held-out bins.

departure described in section A.4, largest just above threshold and for the most cancelling spectra. Figure 4 shows three test spectra, one for each degree of cancellation.

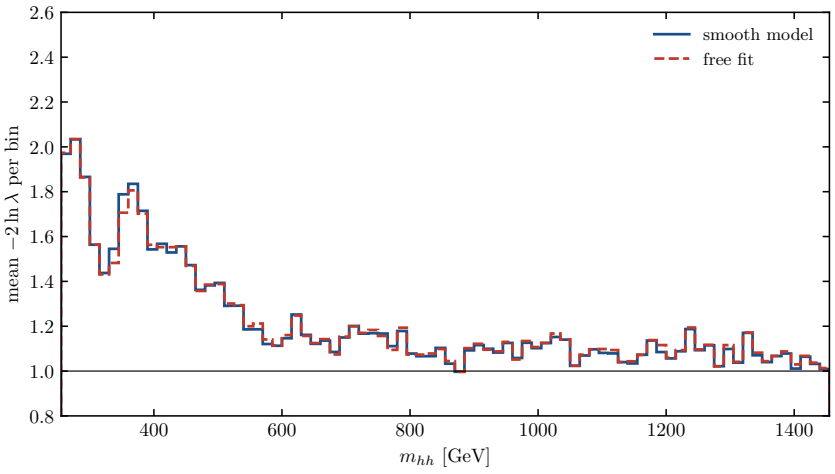


Figure 3: Mean likelihood-ratio statistic $-2\ln\lambda$ per bin on the test spectra as a function of m_{hh}, for the smooth model and the free fit; 1 for a perfect model.

Calibration. With the model-error term the pulls have unit standard deviation within 1%, overall, below and above 555 GeV, and at every degree of cancellation (table 2, figure 2). The total predictive uncertainty is a median 1.014 times the Poisson noise of a POWHEG bin (1.08 in the most cancelling fifth). The pulls of the two lowest bins, 255–285 GeV, where the spectra rise most steeply, have slightly larger standard deviations (1.11 and 1.15).

Inside a bin. To test the model as a density, we fitted it to the training spectra merged into 30 GeV bins (pairs of 15 GeV bins, whose counts add) and predicted the 15 GeV bins of the test spectra. The measure is the χ^2 of the difference between the two halves of each pair, in units of its Poisson noise. Above about 360 GeV the model predicts the split within the noise (χ^2 per pair 1.17 on average between 360 and 555 GeV, at most 1.31; figure 5), as well as the model fitted to the 15 GeV bins does; the free fit, which can only give each half the same value, gives

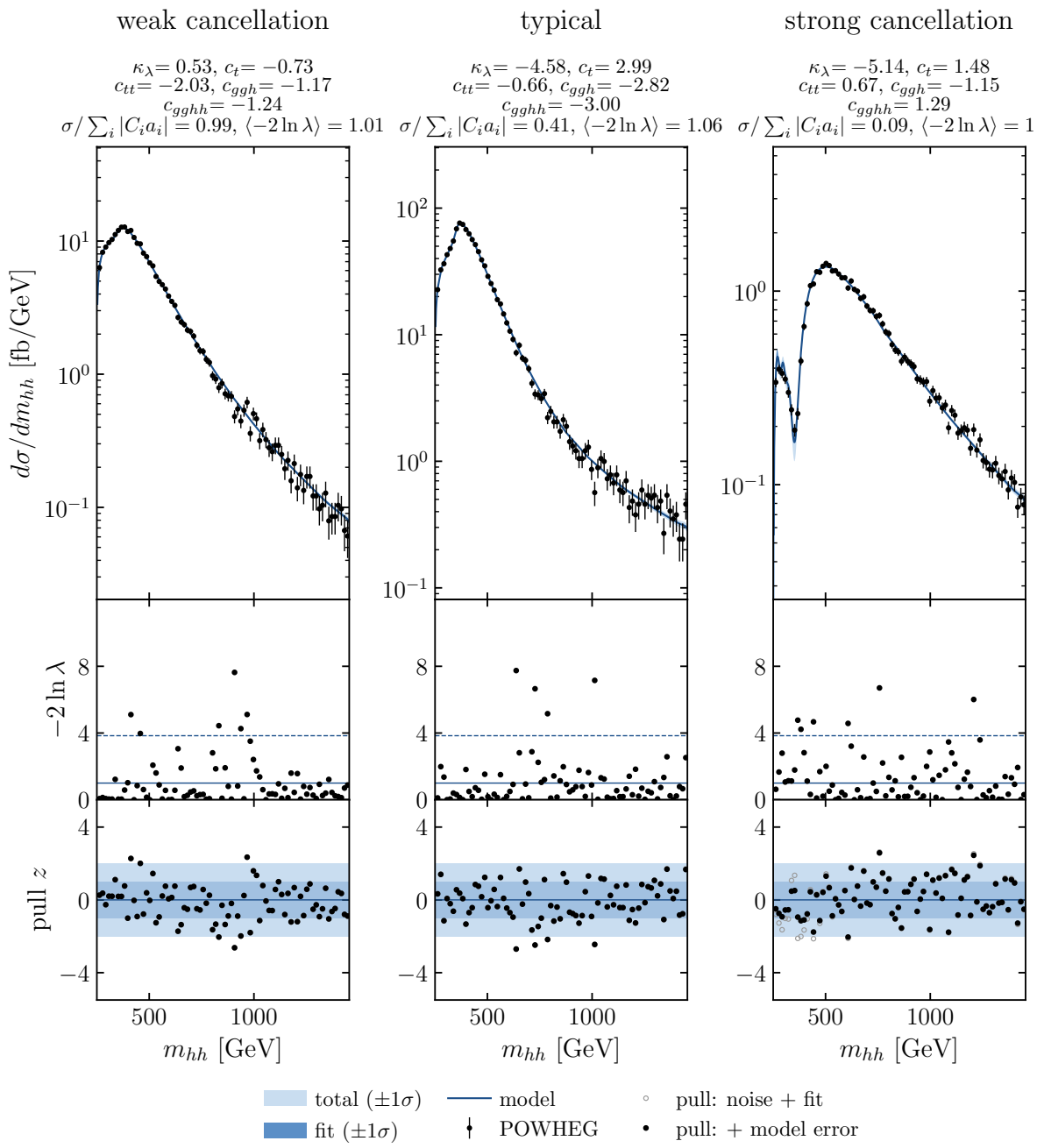


Figure 4: Three of the 1,000 test spectra, chosen by how strongly their 23 terms cancel: weak ($R \approx 1$), typical and strong; in each case the spectrum with the median $-2 \ln \lambda$ within its fifth of the cancellation distribution. Top: POWHEG (points) and the model with its fit and total ($\pm 1\sigma$) bands. Middle: $-2 \ln \lambda$ per bin, with its mean (1, solid) and 95% point (3.84, dashed) for a correct model. Bottom: the pull with the total uncertainty (filled) and without the model-error term (open).

7.9 on average there and up to 34 just above threshold. In the first three pairs, 255–345 GeV, the χ^2 is 11, 3.9 and 4.1: the threshold rise has structure on scales finer than 30 GeV, which 30 GeV bins do not fully determine.

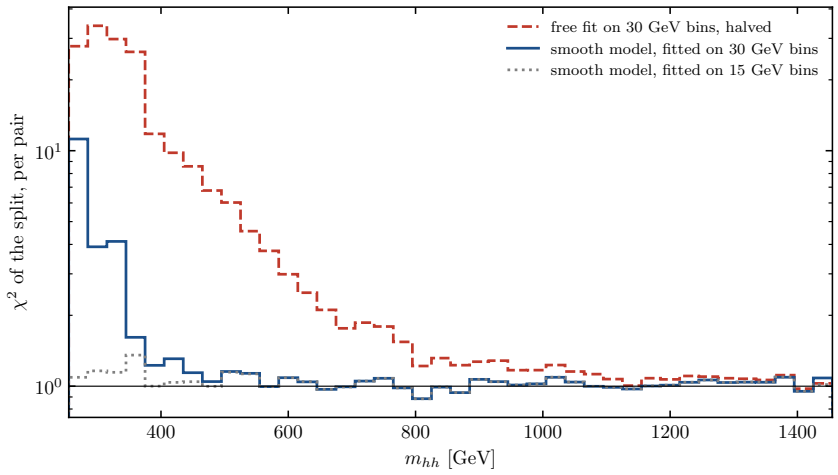


Figure 5: Inside a bin: the model fitted to 30 GeV bins predicts the 15 GeV bins of the test spectra. Shown is the χ^2 per pair of the split of each 30 GeV bin into its two halves, in units of its Poisson noise (1 is perfect), for the smooth model fitted to 30 GeV bins, the free fit (each half given half), and the smooth model fitted to the 15 GeV bins.

The edges. The POWHEG spectra also contain the bins 240–255 and 1455–1500 GeV, not used in the fit. Summed over the test spectra, the model predicts the first 7% high (with the factor ϕ; 55% high without it). Above 1455 GeV the splines are fixed by the prior alone and the extrapolation is unreliable. The model should be used on 250–1455 GeV only.

A.6 The coefficient functions

Figure 6 shows the 23 functions with their exact $\pm 1\sigma$ bands. They are smooth and defined at every m_{hh}. The functions whose monomials are small or nearly degenerate over the sampled θ, a_{16}–a_{20}, are the least well determined (median relative uncertainty 21–74%); their uncertainties are exact and propagate correctly into every prediction.

A.7 Using the model

The model consists of the breakpoints, the posterior mean $\hat{\boldsymbol{\beta}}$ and covariance H^{-1}, the 23 smoothness strengths and the three parameters of $\varepsilon(m)$, in one file, with a short Python module that evaluates it; it requires only NumPy and SciPy [10]. For a HEFT point θ and any binning within 250–1455 GeV it returns the cross section in each bin with its fit uncertainty and model-error term. For a single 15 GeV bin the two are added in quadrature. For a sum over many bins the fit uncertainty is obtained exactly from H^{-1}, while the model-error term lies between the limits of independent bins, $(\sum_k s^2_{\mathrm{model},k})^{1/2}$, and fully correlated bins, $\sum_k s_{\mathrm{model},k}$. For the SM in 255–1455 GeV, for example, $\sigma = 28.11$ fb, with a fit uncertainty of 0.006 fb and a model-error term between 0.03 and 0.17 fb. The density itself, on scales finer than a 15 GeV bin just above threshold, is less certain than its band suggests (appendix B.4).

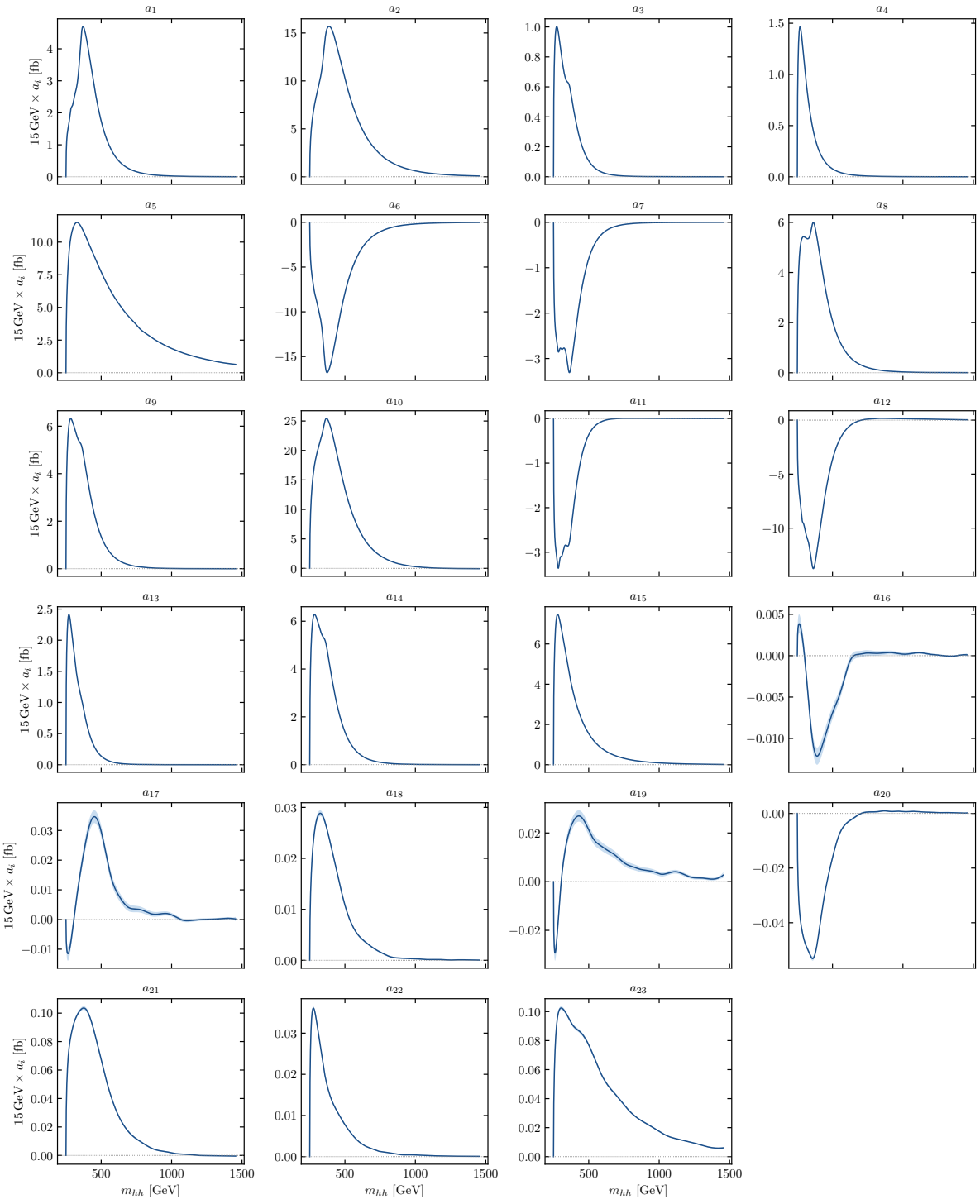


Figure 6: The 23 coefficient functions $a_i(m_{hh})$, shown as $15\,\mathrm{GeV} \times a_i$ (the cross section per 15 GeV multiplying the monomial $C_i(\theta)$ in each panel), with their exact $\pm 1\sigma$ bands. Units are fb.

B A neural-network model of the density

To test whether the results of appendix A depend on the choice of splines and prior, we built a second model of the same 23 coefficient functions in an entirely different way, as a neural network. It uses the same form (1), the same data, the same measures of accuracy and calibration, and, like the smooth model, it is compared with POWHEG through bin integrals. This appendix describes only what differs.

B.1 The network

The 23 coefficient functions are densities in m_{hh}, given by

$$a_i(m) = \phi(m)\, P_i(z)\, e^{Q_i(z)}, \qquad z = 2\sqrt{\frac{m - 2m_h}{1455\ \text{GeV} - 2m_h}} - 1, \tag{7}$$

with the threshold factor ϕ of eq. (2). P is a fully connected network with one input, six hidden layers of 25 units and 23 outputs, with SiLU activations, $x/(1+e^{-x})$ [11, 12], and a tanh on the outputs; Q is a single linear layer. The factor e^{Q_i} sets the overall size of each function, which differs by orders of magnitude between functions, and $P_i \in (-1, 1)$ its shape and sign. The network has 3,944 parameters. Two choices matter. The input z, like the spline breakpoints of appendix A, gives the region 255–435 GeV, where the functions change fastest, a third of its range; with an input linear in m_{hh} that region occupies 12% of the range, and a network of this size does not fit it. A smooth activation makes the density smooth, where ReLU activations would make each a_i piecewise linear in z.

B.2 Bin integrals by Gauss–Legendre quadrature

As for the smooth model, the prediction for bin k is the integral of the density over the bin, $f_k(\theta) = \sum_i C_i(\theta) A_{ik}$ with $A_{ik} = \int_{\text{bin } k} a_i\, dm$. We compute A_{ik} at every training iteration by Gauss–Legendre quadrature with $n = 8$ nodes per bin,

$$A_{ik} \simeq \sum_{g=1}^{n} w_{kg}\, a_i(m_{kg}), \tag{8}$$

which is exact for polynomials of degree up to $2n-1 = 15$. The nodes and weights are computed once, as the eigenvalues and eigenvector components of a tridiagonal matrix [13]. The accuracy of the quadrature matters because the loss sees only the sums (8): a quadrature error would not appear in the fit, but as a bias in the learned density. With eight nodes the relative error of a bin integral is below 3×10^{-5}, against 8×10^{-4} for the model-error coefficient ε at threshold; the trapezoid rule would need a spacing of about 1 GeV near threshold to reach 10^{-3}. Because the a_i depend on m only, the network is evaluated at the 640 fixed nodes once per iteration, whatever the number of training rows in the batch, and each row only selects row k of the table A_{ik}.

B.3 Training

The network is trained on the bins used for fitting in appendix A.1, apart from a small validation set, in two stages, each a fixed number of iterations of the Adam optimiser [14].

1. The absolute difference $|t - f|$, averaged over mini-batches of 2,048 rows, for 200,000 iterations, with a learning rate of 2×10^{-3} halved every 50,000 iterations.

2. The Huber function [15] (with threshold 2) of the pulls $(f-t)/s$, averaged over all training rows, for 5,000 iterations, with a learning rate of 10^{-4} halved every 1,250 iterations. The

variance s^2 is the Poisson variance of the network's own prediction, held fixed within each iteration, so that, as in appendix A.3, the fit is not biased low. The parameters with the lowest loss on a small validation set are kept.

In the second stage a smoothness penalty λR is added to the loss, with

$$R = \frac{1}{23}\sum_{i=1}^{23} \frac{\sum_j \left(g_i''(z_j)\right)^2}{\sum_j g_i(z_j)^2}, \qquad g_i = P_i\, e^{Q_i}, \tag{9}$$

the relative curvature of the network's part of each function (without the threshold factor), evaluated by second differences on 1,201 equally spaced values of z, the variable in which the splines of appendix A are uniformly spaced. The data constrain the poorly determined functions a_{16}–a_{19} weakly, and without the penalty the network fills that freedom with small-scale oscillations: their curvature was 10 to 300 times that of the smooth model's. With $\lambda = 10^{-6}$ it falls to the level of the smooth model's, the curvature of a_1–a_{15} is unchanged, and the accuracy improves slightly; larger values make the network smoother than the smooth model and, at $\lambda = 10^{-4}$, less accurate.

B.4 Results

	mean $-2\ln\lambda$ per bin		median
	255–555 GeV	$>$ 555 GeV	per spectrum
smooth model (appendix A)	1.566	1.100	1.078
free fit (benchmark)	1.559	1.105	1.075
neural-network density model	1.604	1.102	1.073

Table 3: Accuracy of the two models on the 1,000 test spectra (cf. table 1).

	s.d. of the pull				$\langle z^2\rangle$ by fifth of cancellation
model-error term	all	255–555	$>$ 555 GeV	$\lvert z\rvert < 2$	(strongest … weakest)
none	1.104	1.266	1.044	93.9%	1.83, 1.13, 1.07, 1.05, 1.01
its own $\varepsilon(m)$	1.007	1.015	1.005	95.4%	1.00, 0.99, 1.04, 1.04, 1.00
the smooth model's $\varepsilon(m)$	1.003	1.008	1.001	95.5%	0.98, 0.98, 1.03, 1.04, 1.00

Table 4: Calibration of the neural-network density model on the test spectra (cf. table 2).

Accuracy. The network is as accurate as the smooth model above 555 GeV and in the median per spectrum, and slightly less accurate just above threshold (table 3, figure 7).

Calibration and the model-error term. The network has no fit uncertainty of its own; its pulls are computed with the Poisson noise and a model-error term of the form (6), with A_{ik} from the network and $\varepsilon(m)$ determined on the held-out bins exactly as in appendix A.4. Its $\varepsilon(m)$ agrees with the smooth model's from threshold to about 1 TeV (0.070% and 0.077% at 262 GeV; 0.275% and 0.279% at 600 GeV), and with either the pulls are calibrated at every degree of cancellation (table 4, figure 8). Two models built in entirely different ways need the same model-error term: it is a property of the POWHEG spectra, not of either model. Above 1 TeV, where the bins hold about 20 events, the network's ε is lower (2.9% against 3.7% at 1.45 TeV).

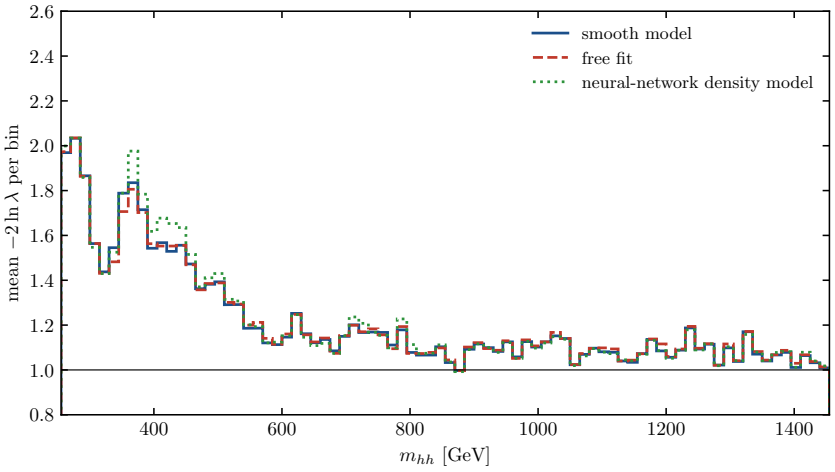


Figure 7: Mean $-2\ln\lambda$ per bin on the test spectra as a function of m_{hh}, for the smooth model, the free fit and the neural-network density model.

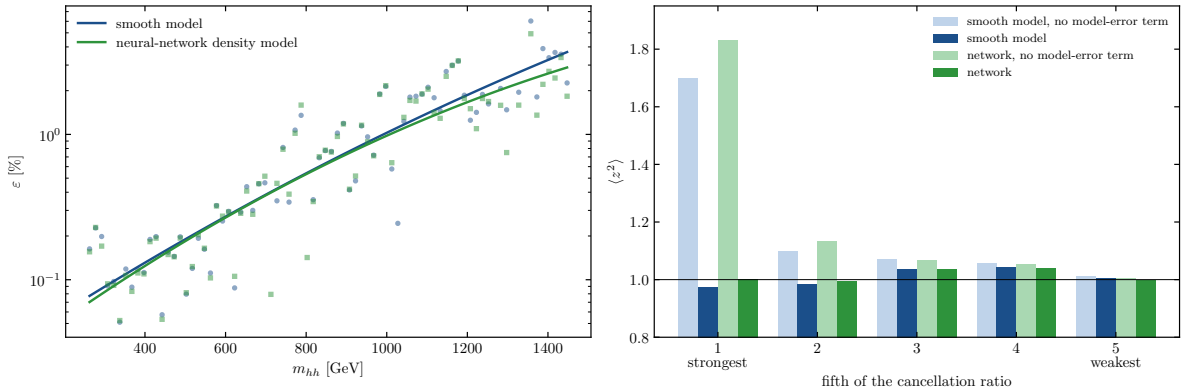


Figure 8: Left: the model-error coefficient $\varepsilon(m_{hh})$ of the two models, per bin (points) and in its smooth form (lines). Right: $\langle z^2 \rangle$ of the pulls by fifth of the cancellation ratio, with and without the model-error term.

The density. The two models can be compared at any m_{hh}, not only through bin integrals. On a 0.5 GeV grid from 255 to 1455 GeV, at the 1,000 test points, the network's $d\sigma/dm_{hh}$ differs from the smooth model's by a median of 0.31% (0.19% below 555 GeV), and equally at the bin centres (0.30%) and between them (0.32%); 90% of the differences lie within the smooth model's total $\pm1\sigma$ (fit uncertainty and model-error term) and 98% within $\pm2\sigma$. Just above threshold the agreement is weaker. Below 350 GeV the median difference is 0.41%, comparable to the smooth model's total uncertainty there (a median 0.54% of the density), and 85% of the differences lie within $\pm2\sigma$. There the differences oscillate on the scale of the bin width while the bin integrals agree (figure 9): the data determine the bin integrals but not the shape of the density on scales finer than a bin, and each model fills in that shape according to its own smoothness. On such scales just above threshold, the difference between the two models is a better measure of the uncertainty of the density than the smooth model's band. The coefficient functions agree with those of the smooth model to 0.04–0.5% of their size for a_1–a_{15} and 1.0–1.3% for a_{20}–a_{23}; the poorly determined a_{16}–a_{19} differ by 2–8%, well within the smooth model's own uncertainties for them (figure 11).

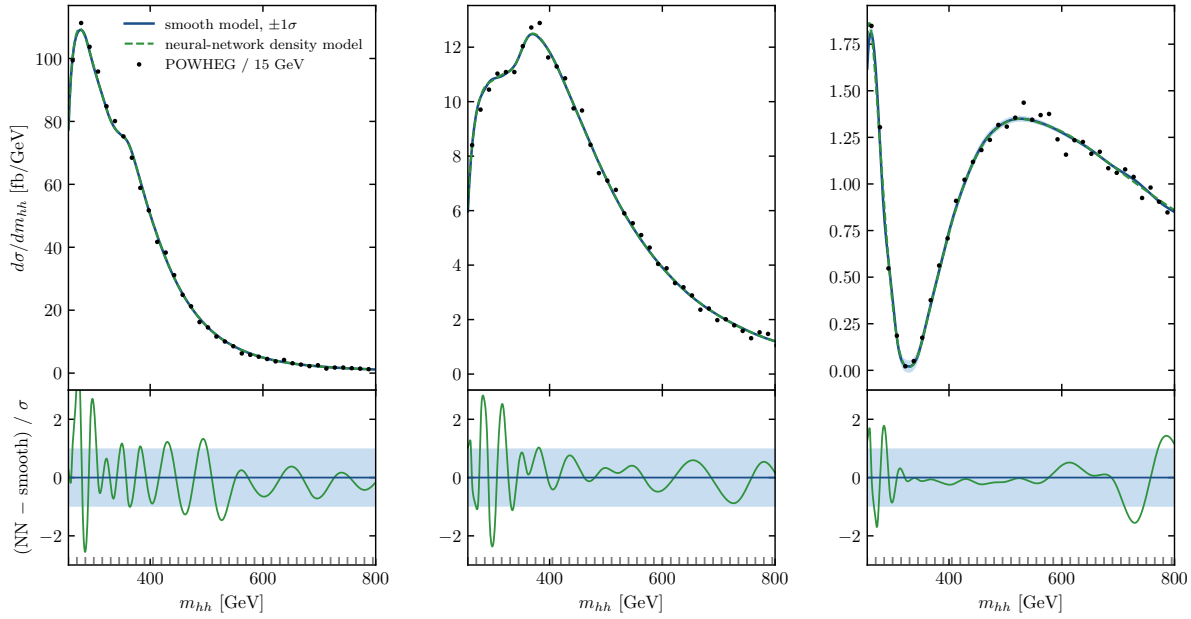


Figure 9: The density $d\sigma/dm_{hh}$ of the two models below 800 GeV for three test spectra, with POWHEG divided by the bin width. Lower panels: the difference between the models in units of the smooth model's total uncertainty; the ticks mark the bin edges.

Inside a bin. Trained on the 30 GeV bins of appendix A.5 with the same procedure, the network predicts the split of each 30 GeV bin into its two 15 GeV halves at least as well as the smooth model (figure 10): the χ^2 per pair is 1.06 on average between 360 and 555 GeV (smooth model 1.17), and 4.7, 2.7 and 3.3 in the first three pairs (smooth model 11, 3.9 and 4.1). Trained on the 15 GeV bins, the two models predict the split equally well.

Below the fitted range. Summed over the test spectra, the network predicts the unused bin 240–255 GeV 7% high, as the smooth model does.

Limitations. The network is a single fit: it has no fit uncertainty of its own, which an ensemble of networks trained from different random initial values would provide. Below 555 GeV its

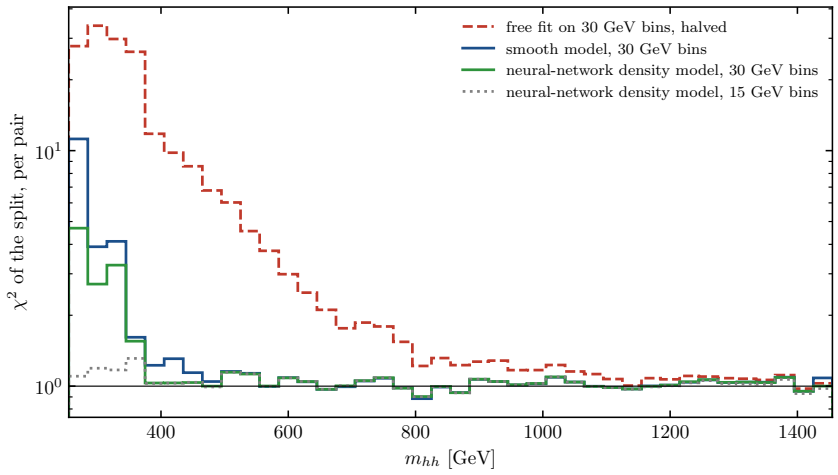


Figure 10: The merged-bin test of figure 5, with the neural-network density model trained on the 30 GeV bins.

accuracy is slightly below that of the smooth model. For these reasons the smooth model is the model of this paper, and the network an independent check of it.

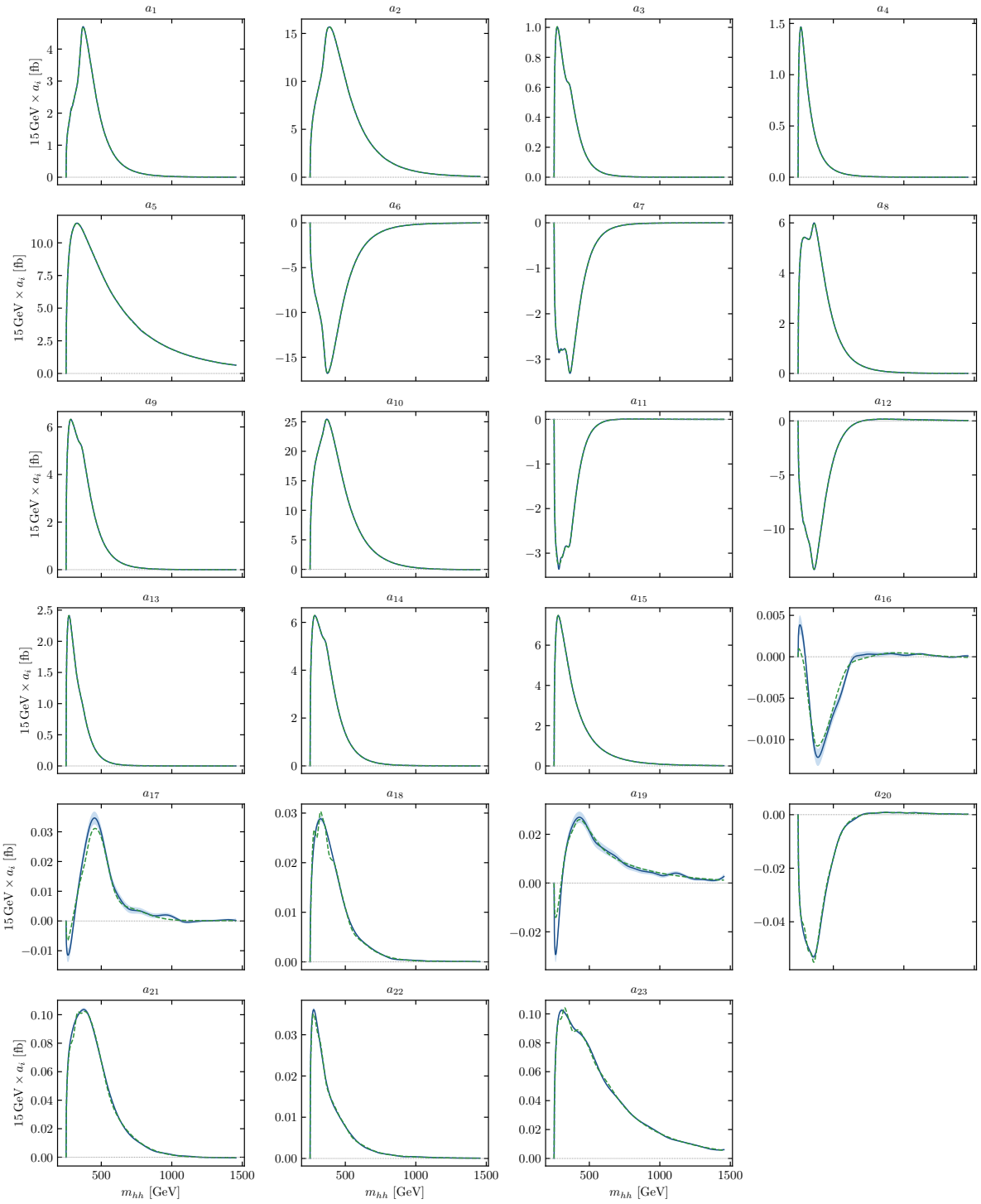


Figure 11: The 23 coefficient functions of the neural-network density model (dashed) over those of the smooth model with their $\pm 1\sigma$ bands, as in figure 6.

References

[1] G. Buchalla, M. Capozi, A. Celis, G. Heinrich, and L. Scyboz. Higgs boson pair production in non-linear Effective Field Theory with full m_t-dependence at NLO QCD. *JHEP*, 09:057, 2018. doi: 10.1007/JHEP09(2018)057.

[2] L. Alasfar et al. Effective Field Theory descriptions of Higgs boson pair production. 2023. LHC Higgs Working Group report.

[3] C. de Boor. *A Practical Guide to Splines*. Springer, New York, 1978.

[4] S. Baker and R. D. Cousins. Clarification of the use of chi-square and likelihood functions in fits to histograms. *Nucl. Instrum. Meth.*, 221:437–442, 1984. doi: 10.1016/0167-5087(84)90016-4.

[5] P. J. Green. Iteratively reweighted least squares for maximum likelihood estimation, and some robust and resistant alternatives. *J. R. Statist. Soc. B*, 46:149–192, 1984.

[6] P. H. C. Eilers and B. D. Marx. Flexible smoothing with B-splines and penalties. *Statist. Sci.*, 11:89–121, 1996. doi: 10.1214/ss/1038425655.

[7] D. J. C. MacKay. Bayesian interpolation. *Neural Comput.*, 4:415–447, 1992. doi: 10.1162/neco.1992.4.3.415.

[8] S. S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *Ann. Math. Statist.*, 9:60–62, 1938. doi: 10.1214/aoms/1177732360.

[9] M. C. Kennedy and A. O'Hagan. Bayesian calibration of computer models. *J. R. Statist. Soc. B*, 63:425–464, 2001. doi: 10.1111/1467-9868.00294.

[10] P. Bhat, N. De Filippis, K. F. Johnson, H. B. Prosper, and R. Starnes. dihiggs: models of the HEFT di-Higgs mass spectrum. `https://github.com/hbprosper/dihiggs`, 2026.

[11] S. Elfwing, E. Uchibe, and K. Doya. Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. *Neural Networks*, 107:3–11, 2018. doi: 10.1016/j.neunet.2017.12.012.

[12] D. Hendrycks and K. Gimpel. Gaussian Error Linear Units (GELUs), 2016.

[13] G. H. Golub and J. H. Welsch. Calculation of Gauss quadrature rules. *Math. Comp.*, 23:221–230, 1969. doi: 10.1090/S0025-5718-69-99647-1.

[14] D. P. Kingma and J. Ba. Adam: A Method for Stochastic Optimization, 2014.

[15] P. J. Huber. Robust estimation of a location parameter. *Ann. Math. Statist.*, 35:73–101, 1964. doi: 10.1214/aoms/1177703732.